

MATHEMATICAL MODELING OF BENT AMMONIA–ALUMINUM AXIAL GROOVED HEAT PIPES FOR SPACECRAFT APPLICATIONS

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A mathematical model for the prediction of the heat transfer characteristics of aluminum–ammonia axial grooved heat pipes with bends is presented. The proposed model accounts for the effects of liquid–vapor interfacial shear stress, the axial variation of the radius of the liquid–vapor interface, and the liquid–solid contact angle between the working fluid and the heat pipe body. The effect of bending is implemented by accounting for the loss of pressure head in both the liquid and the vapor phases of the working fluid due to the presence of the bend. This pressure head loss changes the mass flow rates and the distribution of the working fluid across the length of the heat pipe, which in turn affects the heat transfer characteristics of the heat pipe. The model is validated using experimental data from past studies. A total of 63 different variations of the heat pipe's operating temperature along with the bend angle are studied for this analysis. Based on the observations, a decrease in value is observed in the maximum heat transfer rate (30% decrease in the Q_{max}) and the effective thermal resistance (0.2% decrease in the R_{Total}) as the bend angle is increased from 0° to 180° at a –20°C operating temperature. Plausible reasons for these trends are discussed by analyzing the variations observed in the working fluid physical properties along the length of the heat pipe during its steady state.

KEY WORDS: bent heat pipes, axial grooved heat pipes, satellite thermal control, mathematical modeling

1. INTRODUCTION AND LITERATURE REVIEW

Heat pipes are one of the most widely used heat transfer devices for spacecraft thermal management. They are generally used to transfer heat from detectors, electronic subsystems, and other heat-dissipating components within the payload to the spacecraft's radiator, which is responsible for pumping out heat into interstellar space.

Owing to the operating temperature requirements of the electronics and the radiator, ammonia heat pipes serve as a great appliance for transferring heat at temperatures ranging from –60°C to 100°C. This is because the value of the figure of merit of ammonia as a working fluid in heat pipes is the highest in the given temperature range. These heat pipes are made of aluminum for two reasons: one is the excellent compatibility of aluminum with ammonia (Pratt, 1976; Cook et al., 1961), which ensures the longevity of the device, and another is that no new compounds are formed that could disrupt its operation in the long run. The other benefit of using aluminum is that it is lightweight, which helps cut down on the payload and enables engineers to use that extra weight allowance to optimize the system for the most efficient working at the optimum weight. For more than 40 years, aluminum–ammonia heat pipes have been a proven technology for spacecraft thermal control, as they can transport heat over long distances (3 m or more) from a heat source to a heat sink with a very small temperature difference (Ababneh et al., 2018).

NOMENCLATURE

Symbols

D	diameter, m
A	area, m ²
g	acceleration due to gravity (9.81), m s ⁻²
N	total number of grooves, dimensionless
h	height, m
Q	heat, W
m	mass, kg
f	friction factor, dimensionless
V	volume, m ³
L	length, m
$\dot{m}$	mass flow rate, kg s ⁻¹
R	radius, m
Re	Reynolds number, dimensionless
R_g	gas constant, J kg ⁻¹ K ⁻¹
w	width of the groove, m
x	coordinate, m
j	iteration number, dimensionless
p	pressure, Pa
u	velocity, m s ⁻¹
T	temperature, K
k	thermal conductivity, W m ⁻¹ K ⁻¹
t	wall thickness, m
fRe	Poiseuille number
R_1	resistance of the metal container across the evaporator
R_2	resistance of the metal fins in the evaporator
R_3	thin liquid film region resistance across the evaporator section
R_4	resistance of the metal wall (thickness) of the evaporator section
R_5	working fluid thermal resistance across the evaporator section
R_6	thin liquid film region resistance across the condenser section
R_7	resistance of the metal fins in the condenser
R_8	resistance of the metal container across the condenser
R_9	working fluid thermal resistance across the condenser section

R_{10}	resistance of the metal wall (thickness) of the condenser section
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Greek Symbols

β	angle of bend, degrees
ν	kinematic viscosity, m ² s ⁻¹
μ	absolute viscosity, Pa s
λ	latent heat of vaporization, J kg ⁻¹
χ	liquid–vapor interfacial shear stress factor, (dimensionless)
Δ	difference
σ	surface tension, N m ⁻¹
ρ	density, kg m ⁻³
α	contact angle, degrees
γ	half groove angle, degrees
δ	liquid thickness, m
ψ	liquid meniscus radius angle
ϕ	vapor radius angle
ω_0	outer width of the groove, mm
ω_1	upper width of the groove, mm
ω_2	bottom width of the groove, mm
2υ	inclination angle of the groove, degrees

Abbreviations and subscripts

cond	condenser
evap	evaporator
adia	adiabatic
ISRO	Indian Space Research Organization
AGHP	axial grooved heat pipe
l	liquid
meni	liquid–vapor meniscus
hyd	hydraulic
max	maximum
g	groove
in	input
c	capillary, condenser section
e	evaporator section
a	adiabatic section
fin	fin of groove
out	output
v	vapor
wall	heat pipe wall
film	liquid film

Khrustalev and Faghri (1995) developed a one-dimensional model describing the fluid flow and heat and mass transfer in a miniature axial rectangular grooved heat pipe to predict the maximum heat transfer restricted by both capillary limitation and the incipience of nucleate boiling. Hopkins et al. (1999) carried out experimentation with several flat miniature heat pipes to study the maximum heat flow rates and heat fluxes in them and developed a mathematical model to predict the behavior of the investigated heat pipes. Do et al. (2006) developed a mathematical model to predict the thermal performance of flat microheat pipes by accounting for the liquid–vapor interfacial shear stress, the solid–liquid contact angle, and the amount of liquid charge inside the heat pipe, and successfully predicted the experimental results reported by Hopkins et al. (1999).

By deriving a validated one-dimensional mathematical model based on a semi-analytic hydraulic approach, Arab and Abbas (2014) developed a reduced-order mathematical model to predict the performance of axial grooved heat pipes (AGHPs), which was followed by the derivation of reduced-order model parameters using experimental results. In this study, they calculated the thermal resistances of the trapezoidal grooves and used them to predict the thermal performance of several working fluids. By taking into consideration the axial variation of the liquid–vapor interfacial radius and the effects of liquid–vapor interfacial stress, Anand (2019) developed an analytical model for predicting the heat transport capability of an AGHP. The model was then utilized to study the effects of charge quantity on the nominal charge quantity of the heat pipe. Desai et al. (2019) developed a numerical model for AGHPs to predict the maximum heat transfer rate and the total thermal resistance of the heat pipe, which was experimentally validated, followed by the conduction of a parametric study to study the effects of the geometrical parameters of the axial grooves in the wick of the heat pipe.

In space applications, clusters of electronics packed compactly to utilize all the space available comprises the payload. Hence, generally, the source of heat generation and the heat sink (radiator) do not lie in the same plane or have several components between them; therefore, joining a conventional straight heat pipe may not be feasible. In these cases, the conventional heat pipes are bent to be able to connect the sources and the sink. This is done at a fraction of the heat transport capability of the heat pipe. This is possible due to the loss of pressure head (and hence loss of energy) at the bend due to factors such as loss of momentum and the formation of eddies in the flow of the working fluid (Rowe, 1970; Babcock & Wilcox Company, 1978; Idelchik, 1986). To the best of the author's knowledge, no study is available in the open literature that deals with the effect of bend on the thermal performance of AGHPs. Therefore, this paper is an attempt to predict the variation of the maximum heat transfer rate (Q_{max}) and the total thermal resistance (R_{Total}) in bent AGHPs upon varying their bend angle. A numerical model is developed, working on the principle of limiting the contact angle (Do et al., 2008) between the solid heat pipe body and the liquid–vapor interfacial film meniscus due to the variation of this film's curvature. The numerical model is validated with the experimental results obtained by Desai et al. (2019) and Anand (2019).

To evaluate the effect of bending, the loss of pressure head occurring in the liquid and vapor phases of the working fluid due to the presence of a bend is accounted for using empirical relations and formulae derived from the analysis of steam pipes with bends (Rowe, 1970; Babcock & Wilcox Company, 1978; Idelchik, 1986; Chisholm, 1980). By doing so, the variations in Q_{max} and R_{Total} are accurately predicted for aluminum-based bent axial grooved heat pipes charged with ammonia at seven different bend angles (from 0° to 180°) over the entire operating temperature range (from –60°C to 100°C) of these heat pipes under microgravity conditions.

2. MATHEMATICAL MODEL

For the current study, the chosen combination of heat pipe material and working fluid is ammonia–aluminum due to their excellent compatibility and the high figure of merit of ammonia in the range of operating temperatures of a typical satellite application.

2.1 Assumptions

The model has been developed under the following assumptions:

- The heat pipe is assumed to be isothermal, implying constant physical properties of the fluid throughout the length of the heat pipe.

- The working fluid fully saturates the heat pipe grooves.
- Steady-state condition for heat transfer and fluid flow is present.
- Liquid and vapor flows are unidirectional and incompressible.
- The limiting contact angle is assumed to be constant at all operating temperatures.
- Bending has a negligible effect on the groove and vapor core dimensions.

2.2 Governing Equations

The capillary pressure developing inside the axial grooves of the AGHP is what fundamentally drives its functioning. The surface tension that enables this phenomenon primarily acts to pull the liquid inside the axial grooves. It does this by pulling the liquid–vapor interfacial meniscus where it meets the heat pipe’s solid surface (on the groove fins). The Young–Laplace equation then enables us to calculate the pressure difference across the liquid–vapor interface. By running a differential-integral-algebraic model based on this equation, it is possible to predict the capillary pressure acting inside the heat pipe, which thus enables us to predict how the working fluid behaves inside the heat pipe in both phases and to determine how the heat pipe will perform during operation. Figure 1 shows the schematic of a heat pipe, along with the coordinate system used for the mathematical modeling.

The pressure difference between the liquid and vapor phases of an AGHP at a position x along the heat pipe axis can be given by the Young–Laplace equation (Khrustalev and Faghri, 1995):

$$P_v(x) - P_l(x) = \frac{\sigma}{R_c(x)} \quad (1)$$

R_c denotes the radius of curvature of the liquid–vapor meniscus at x . Upon differentiating both sides by x , we get

$$\frac{dP_v}{dx} - \frac{dP_l}{dx} = \left(-\frac{\sigma}{R_c^2(x)} \right) \cdot \frac{dR_c(x)}{dx} \quad (2)$$

Here, the pressure gradient of the liquid is stated as

$$\frac{dP_l}{dx} = \left(-\frac{f_l \cdot \rho_l \cdot (u_l)^2}{2 \cdot D_{l,hyd}} \right) \cdot \chi \quad (3)$$

In the above formula, χ is known as the liquid–vapor interfacial shear stress factor (Schneider, 1980; Anand, 2019) and is defined as

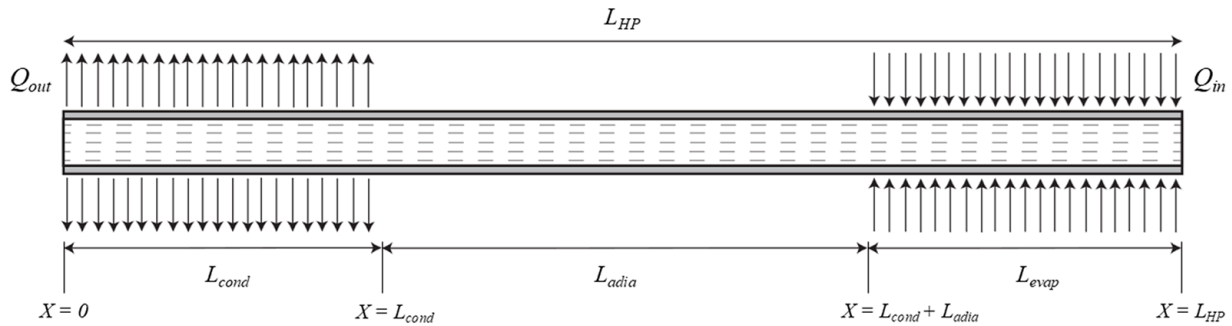


FIG. 1: Heat pipe schematic diagram, along with the coordinate system

$$\chi = \frac{f_l Re_l}{(f_l Re_l)_0} = \left[1 + \frac{64 \cdot N \cdot (\omega_1)^3}{24 \cdot \pi \cdot (D_{l,hyd})^3} \cdot \frac{v_v}{v_l} \cdot \left[1 - 1.1971 \cdot e^{\left[-\frac{\pi \cdot h_g}{\omega_1} \right]} \right] \right] \quad (4)$$

It is the ratio of $f_l Re_l$ (the Poiseuille's number for liquid flow with the interfacial shear stress) and $(f_l Re_l)_0$ (the Poiseuille's number for liquid flow without the interfacial shear stress).

For turbulent and laminar liquid flows, the friction factor (f_l) can be given as

$$\chi = \begin{cases} \frac{64}{Re_l}, & Re_l \leq 2000 \\ \frac{0.316}{Re_l^{0.25}}, & Re_l > 2000 \end{cases} \quad (5)$$

where Re is the Reynold's number for the liquid flow, given by

$$Re_l = \frac{\rho_l \cdot u_l \cdot D_{l,hyd}}{\mu_l} \quad (6)$$

The liquid flow velocity at a given x position is

$$\chi = \begin{cases} \frac{Q \cdot x}{\lambda \cdot \rho_l \cdot A_l \cdot l_c}, & x \leq l_c \\ \frac{Q}{\lambda \cdot \rho_l \cdot A_l}, & x \leq l_c + l_a \\ \frac{Q \cdot (l_{hp} - x)}{\lambda \cdot \rho_l \cdot A_l \cdot l_e}, & l_c + l_a < x \end{cases} \quad (7)$$

The area of the liquid flow at the x position (Arab and Abbas, 2014) is

$$A_l = \left(\frac{\omega_1 + \omega_2}{2} \cdot h_g + \frac{\omega_1 \cdot r \cdot \cos(\psi(r))}{2} - \psi(r) \cdot r^2 \right) \cdot N \quad (8)$$

where ψ is the meniscus radius angle given by

$$\psi = \sin^{-1} \left(\frac{\omega_1}{2 \cdot r} \right) \quad (9)$$

The hydraulic diameter for the axial grooves is given as

$$D_{l,hyd} = \frac{4 \cdot A_l}{N \cdot \left(\omega_2 + 2 \cdot \sqrt{\left(\frac{\omega_2 - \omega_1}{2} \right)^2 + h_g^2} \right)} \quad (10)$$

where N is the number of grooves and ω_1 and ω_2 are the upper and bottom groove widths, respectively. The cross section of an AGHP along with an enlarged view of an axial groove is shown in Fig. 2.

Similarly, the vapor pressure gradient under microgravity conditions in an AGHP can be defined as (Anand, 2019)

$$\frac{dP_v}{dx} = \left(-\frac{f_v \cdot \rho_v \cdot (u_v)^2}{2 \cdot D_{v,hyd}} \right) \quad (11)$$

For laminar and turbulent vapor flows, the friction factor (f_v) can be given as

$$f_v = \begin{cases} \frac{64}{Re_v}, & Re_v \leq 2000 \\ \frac{0.316}{Re_v^{0.25}}, & Re_v > 2000 \end{cases} \quad (12)$$

where Re is the Reynold's number for the vapor flow, given by

$$Re_v = \frac{\rho_v \cdot u_v \cdot D_{v,hyd}}{\mu_v} \quad (13)$$

The vapor flow velocity at a given x position is given by

$$u_v = \begin{cases} \frac{Q \cdot x}{\lambda \cdot \rho_v \cdot A_v \cdot l_c}, & x \leq l_c \\ \frac{Q}{\lambda \cdot \rho_v \cdot A_v}, & x \leq l_c + l_a \\ \frac{Q \cdot (l_{hp} - x)}{\lambda \cdot \rho_v \cdot A_v \cdot l_e}, & l_c + l_a < x \end{cases} \quad (14)$$

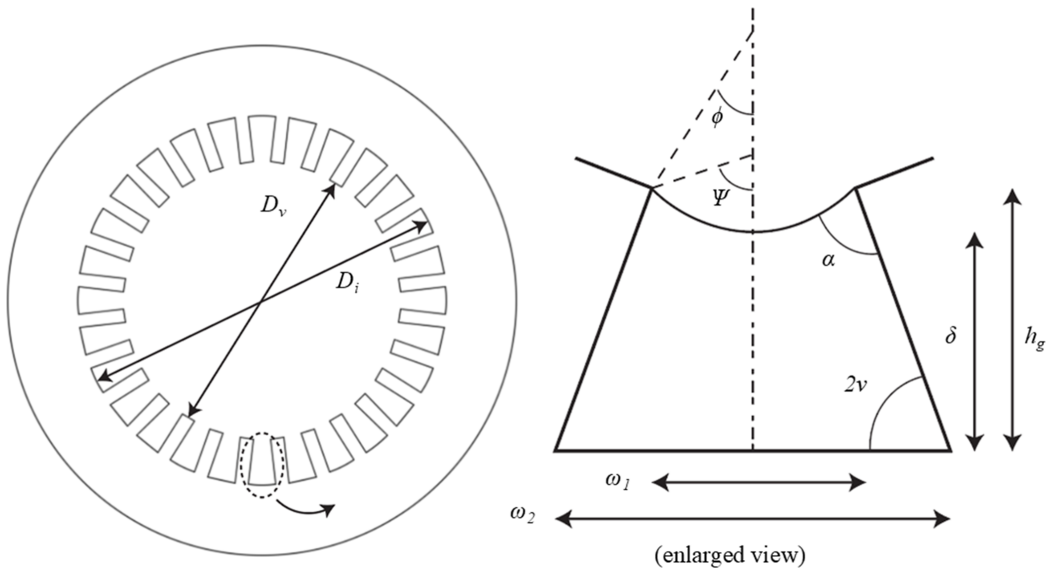


FIG. 2: Schematic diagram of the cross section of heat pipe and axial grooves

where the area of vapor flow will be (Arab and Abbas, 2014)

$$A_v = \frac{\pi \cdot D_v^2}{4} + \left[\psi(r) \cdot r^2 - \frac{D_v^2 \cdot \phi}{4} - \frac{\omega_1}{2} \cdot \cos(\psi(r)) \cdot r + \frac{\omega_1}{2} \cdot \frac{\cos(\phi) \cdot D_v}{2} \right] \cdot N \quad (15)$$

where ϕ is the vapor channel radius angle, given by

$$\phi = \sin^{-1} \left(\frac{\omega_2}{D_v} \right) \quad (16)$$

For vapor flow, the hydraulic diameter is

$$D_{v,hyd} = \frac{4 \cdot A_v}{\pi \cdot D_v + 2 \cdot N \cdot r \cdot \psi(r) - N \cdot \phi \cdot D_v} \quad (17)$$

Inside the heat pipe, the axial heat flow can be determined by the position of the point under consideration along the heat pipe axis (as a function of x distance from the condenser end of the heat pipe). This can be explained by understanding how each section of the heat pipe behaves. Inside the evaporator section, the heat load steadily increases as heat is input. Because the adiabatic region is insulated and does not allow for flow of heat through its surface (hence the name adiabatic), the heat load remains constant in it. Furthermore, as we move to the condenser section, the continuous removal of heat enables the heat load to decrease gradually back to zero.

$$Q_{Axial Load} = \begin{cases} \frac{Q \cdot x}{l_c}, & x \leq l_c \\ Q, & x \leq l_c + l_a \\ \frac{Q \cdot (l_{hp} - x)}{l_e}, & l_c + l_a < x \end{cases} \quad (18)$$

2.2.1 Effect of Bends

The effect of bending in a straight heat pipe is implemented by accounting for the loss of pressure head that occurs in bends inside pipes because of the loss of momentum (and hence energy) due to the change in direction, which also results in the formation of centrifugal forces and secondary flows that generate additional turbulence and cause the fluid to slow down (Rowe, 1970; Babcock & Wilcox Company, 1978; Idelchik, 1986).

The loss of pressure head in fluids due to bends in a pipe is given as (Babcock & Wilcox Company, 1978)

$$P_{loss, bend} = \left(\frac{1}{2} \cdot \rho \cdot (u)^2 \cdot \left(\frac{f \cdot R_{bend} \cdot \pi \cdot (\beta)}{D_{hyd} \cdot 180} + k_b \right) \right) \quad (19)$$

When implementing this effect in an AGHP, it will have to be done in both liquid and vapor phases; that is, in the liquid moving in axial grooves as well as the vapor moving in the vapor core through the bend. Since neither the vapor core nor the axial grooves can be considered ideal pipes, we consider their hydraulic diameters at any given x position to regard the changes that develop due to the changing value of the radius of the meniscus of the liquid in the grooves. Since the interaction of liquid and vapor phases across the meniscus film is all determined by the radius of the meniscus in such a numerical model (such that any change in P_l and P_v at any given x will not have a

direct impact on the vapor or liquid flowing on the other side of the meniscus, respectively), considering the value and variations in the radius of meniscus curvature is assumed to be sufficient to implement the changes due to the bend in the operation of the heat pipe. Figure 3 depicts the different variables that define the characteristics of the bend in a heat pipe.

We intend to observe the effect on the performance of the heat pipe caused by bending it at different bend angles. In the current analysis, the radius of bend (R_{bend}) is 10 mm to reduce the number of variables. Bending the pipe at an angle β , the length of the bend can be calculated to be

$$l_{bend} = \frac{R_{bend} \cdot \pi}{180} \quad (20)$$

For our analysis, the bends are assumed to be situated in the middle of the heat pipe (or of the adiabatic section). The liquid pressure head loss gradient due to the bend can be calculated to be (Babcock & Wilcox Company, 1978)

$$\left(\frac{dP_l}{dx} \right)_{loss, bend} = \frac{\frac{1}{2} \cdot \rho_l \cdot (u_l)^2 \cdot \left(\frac{f_l \cdot R_{bend} \cdot \pi \cdot (\beta)}{D_{l,hyd} \cdot 180} + k_b \right)}{l_{bend}} \quad (21)$$

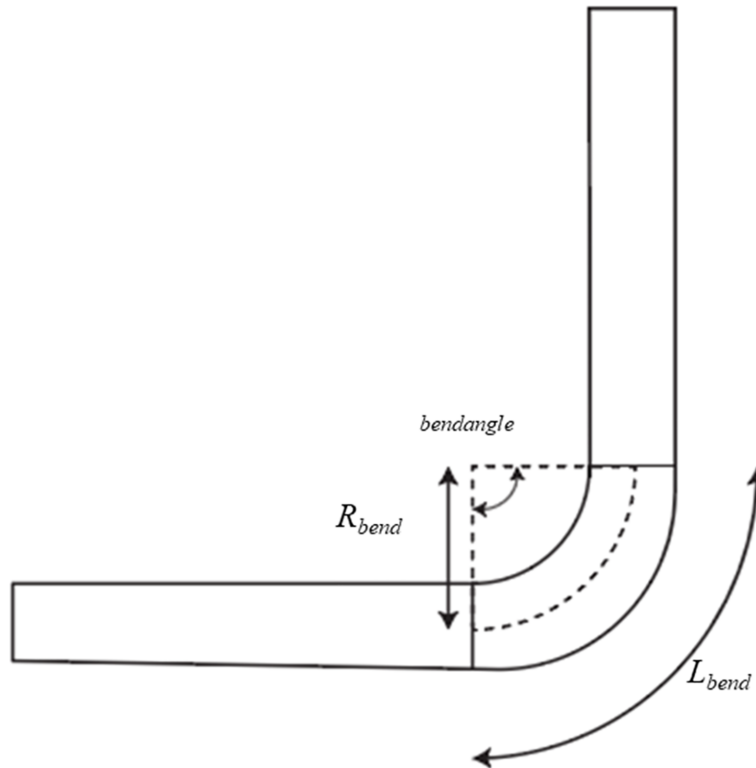


FIG. 3: Nomenclature of bent heat pipes

This pressure loss gradient will augment the traditional pressure gradient throughout the length of the bend. Therefore, the liquid pressure gradient for a bent heat pipe will be given as

$$\frac{dP_l}{dx} = \left(-\frac{f_l \cdot \rho_l \cdot (u_l)^2}{2 \cdot D_{l,hyd}} \right) \cdot \chi - \left(\frac{dP_l}{dx} \right)_{loss} \quad (22)$$

where the augmented liquid pressure gradient loss due to the bend is

$$\left(\frac{dP_l}{dx} \right)_{loss} = \begin{cases} 0, x < \frac{l_{hp}}{2} - \frac{l_{bend}}{2} \\ \frac{\frac{1}{2} \cdot \rho_l \cdot (u_l)^2 \cdot \left(\frac{f_l \cdot R_{bend} \cdot \pi \cdot (\beta)}{D_{l,hyd} \cdot 180} + k_b \right)}{l_{bend}}, x \leq \frac{l_{hp}}{2} + \frac{l_{bend}}{2} \\ 0, \frac{l_{hp}}{2} + \frac{l_{bend}}{2} < x \end{cases} \quad (23)$$

Similarly, the pressure gradient of vapor for a bent heat pipe can be shown as

$$\frac{dP_v}{dx} - \left(\frac{f_v \cdot \rho_v \cdot (u_v)^2}{2 \cdot D_{v,hyd}} \right) + \left(\frac{dP_v}{dx} \right)_{loss} \quad (24)$$

where the augmented vapor pressure gradient loss due to the bend is

$$\left(\frac{dP_v}{dx} \right)_{loss} = \begin{cases} 0, x < \frac{l_{hp}}{2} - \frac{l_{bend}}{2} \\ \frac{\frac{1}{2} \cdot \rho_v \cdot (u_v)^2 \cdot \left(\frac{f_v \cdot R_{bend} \cdot \pi \cdot (\beta)}{D_{v,hyd} \cdot 180} + k_b \right)}{l_{bend}}, x \leq \frac{l_{hp}}{2} + \frac{l_{bend}}{2} \\ 0, \frac{l_{hp}}{2} + \frac{l_{bend}}{2} < x \end{cases} \quad (25)$$

Hence, combining Eqs. (2), (22), and (24), the expanded form of the Young–Laplace equation for a bent AGHP along its axis can be given as

$$\frac{dR_c(x)}{dx} = \frac{R_c^2(x)}{\sigma} \cdot \left[\left[\left(\frac{f_l \cdot \rho_l \cdot (u_l)^2}{2 \cdot D_{l,hyd}} \right) \cdot \chi - \left(\frac{dP_l}{dx} \right)_{loss} \right] - \left[\left(\frac{f_v \cdot \rho_v \cdot (u_v)^2}{2 \cdot D_{v,hyd}} \right) + \left(\frac{dP_v}{dx} \right)_{loss} \right] \right] \quad (26)$$

Along with Q_{max} , the total thermal resistance (R_{total}) is also an important indicator for the performance of any heat pipe. As reported by Arab and Abbas (2014) and Wong et al. (2012), there are essentially two major paths for heat transfer in an AGHP. These paths can be described as (1) the macro region (through the metal container and then through the liquid-saturated wick) and (2) the micro region (through the metal container, metal fins, and then through the liquid film region). Figure 4 shows the various thermal resistances that form these two regions. Path (1) follows

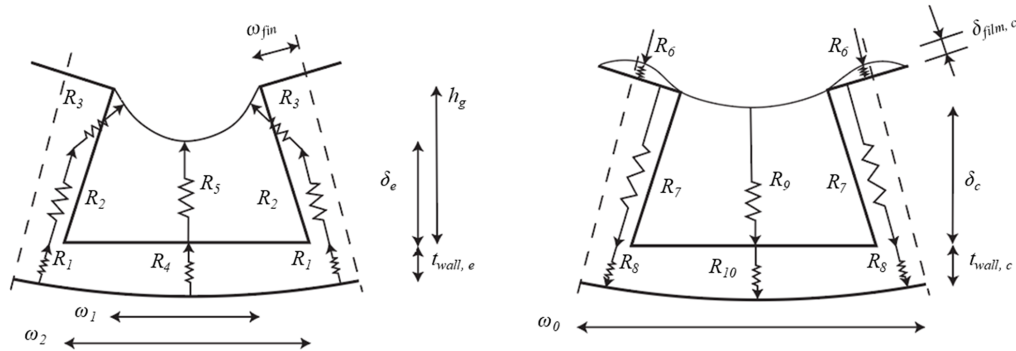


FIG. 4: Groove liquid layer cross section at the evaporator and condenser ends

the thermal resistance network through the metal wall of the heat pipe (consisting of R_4 in the evaporator section and R_{10} in the condenser section), followed by resistance offered by the liquid phase of the working fluid (consisting of R_5 in the evaporator section and R_9 in the condenser section). Path (2), which is in parallel combination (of resistors) with the previous path, goes through the metal container of the heat pipe (namely R_1 in the evaporator section and R_8 in the condenser section), the metal fins (R_2 and R_7 , respectively), followed by a very thin liquid film region inside the grooves (R_3 and R_6 , respectively).

These various thermal resistances can be calculated as (Arab and Abbas, 2014)

$$R_1 = \frac{t_{wall,e}}{k_{pipe}\omega_{fin}} \quad (27)$$

$$R_2 = \frac{\delta}{k_{pipe}\omega_{fin}} \quad (28)$$

$$R_3 = \frac{0.185\omega_{fin}}{k_f(h_g - \delta_e)} \quad (29)$$

$$R_4 = \frac{t_{wall,e}}{k_{pipe}\omega_0} \quad (30)$$

$$R_5 = \frac{\delta_e}{k_f\omega_2} \quad (31)$$

$$R_6 = \frac{\delta_{film,c}}{k_f\omega_{fin}} \quad (32)$$

$$R_7 = \frac{t_{fin}}{k_{pipe}\omega_{fin}} \quad (33)$$

$$R_8 = \frac{t_{wall,c}}{k_{pipe}\omega_{fin}} \quad (34)$$

$$R_9 = \frac{\delta_c}{k_f\omega_1} \quad (35)$$

$$R_{10} = \frac{t_{wall,c}}{k_{pipe}\omega_2} \quad (36)$$

Here, the thickness of the liquid layer in the grooves is given by

$$\delta = h_g - r \cdot (1 - \cos(\psi(r))) \quad (37)$$

The liquid film thickness is given by

$$\delta_{film} = (1.8171) \cdot \left(\frac{\mu_l \omega_{fin} Q r}{l_c N \sigma \rho_l \lambda} \right)^{\frac{1}{3}} \quad (38)$$

Using these values for various thermal resistances across the heat pipe, the net resistances of the three heat pipe sections can be given as

$$R_e = \frac{(R_1 + R_2 + R_3)(R_4 + R_5)}{l_e N (R_1 + R_2 + R_3 + 2(R_4 + R_5))} \quad (39)$$

$$R_a = \frac{T_{v,c} T_{v,e} R_g \log \left(\frac{P_{v,(evap)}}{P_{v,(cond)}} \right)}{\lambda Q} \quad (40)$$

$$R_c = \frac{(R_6 + R_7 + R_8)(R_9 + R_{10})}{l_e N (R_6 + R_7 + R_8 + 2(R_9 + R_{10}))} \quad (41)$$

Hence, the total thermal resistance can be found to be

$$R_{total} = R_a + R_e + R_c \quad (42)$$

2.3 Solution Algorithm

Do et al. (2008) and Wong et al. (2012) reported that the heat pipe operation limiting condition is reached when the liquid in the grooves at the evaporator end forms a contact angle equal to the least possible contact angle for the liq-

uid–solid combination (i.e., the working fluid and the heat pipe wick material). This is explained by the condition that if the angle formed and the solid container and the liquid–vapor film meniscus becomes less than the limiting value, it will result in the rupturing of the film meniscus, which will result in a pressure imbalance within both phases of the working fluid and hence initiate the heat pipe dry-out.

The value of the film meniscus curvature radius will be the least at the evaporator end at any value of Q_{in} throughout the heat pipe length (as shown by Fig. 5). The rate of evaporation will be highest at that location, due to which the groove liquid–layer thickness will be the least.

Hence reducing the meniscus curvature radius value and therefore, the limiting value of $R_{c,min}$, is given by

$$R_{c,min} = \frac{\omega_1}{2 \cos(\alpha_{min})} \quad (43)$$

where α_{min} is the limiting contact angle value for the aluminum–ammonia combination. Mathematically, the difference between ω_1 and ω_2 is neglected to calculate the value of $R_{c,min}$. Based on the experimental analysis of aluminum–ammonia heat pipes conducted at the Space Applications Center (Indian Space Research Organization), the value of α_{min} was reported to be 67° (Desai et al., 2019).

For our mathematical model, an aluminum–ammonia AGHP having 28 trapezoidal grooves is considered. An ID of 10.5 mm is assumed with a vapor-core diameter of 7.9 mm. This gives us a groove height (h_g) of 1.3 mm. The groove top width (ω_1) is 0.5 mm, and the groove bottom width (ω_2) is 0.79 mm. The evaporator and condenser sections are of equal length, 300 mm long, and the adiabatic section is assumed to be 400 mm long.

We obtain the R_c variation across the heat pipe length by solving Eq. (26) with the help of the classical fourth-order Runge–Kutta method. For a value of Q_{in} (heat input), we apply this method assuming R_c ($x = 0$; i.e., the condenser end) is equal to 1 m (or any other high value for a flat meniscus), and solve from $x = 0$ to $x = L_{hp}$ (i.e., 1 m), with a step size (Δx) of 1 mm; hence the number of nodes equal to 1000 (given by $L_{hp}/\Delta x$). After running the model for a Q_{in} , if the value of R_c ($x = 1$ m) is within 0.1% of the $R_{c,min}$, that value of Q_{in} is said to be the Q_{max} .

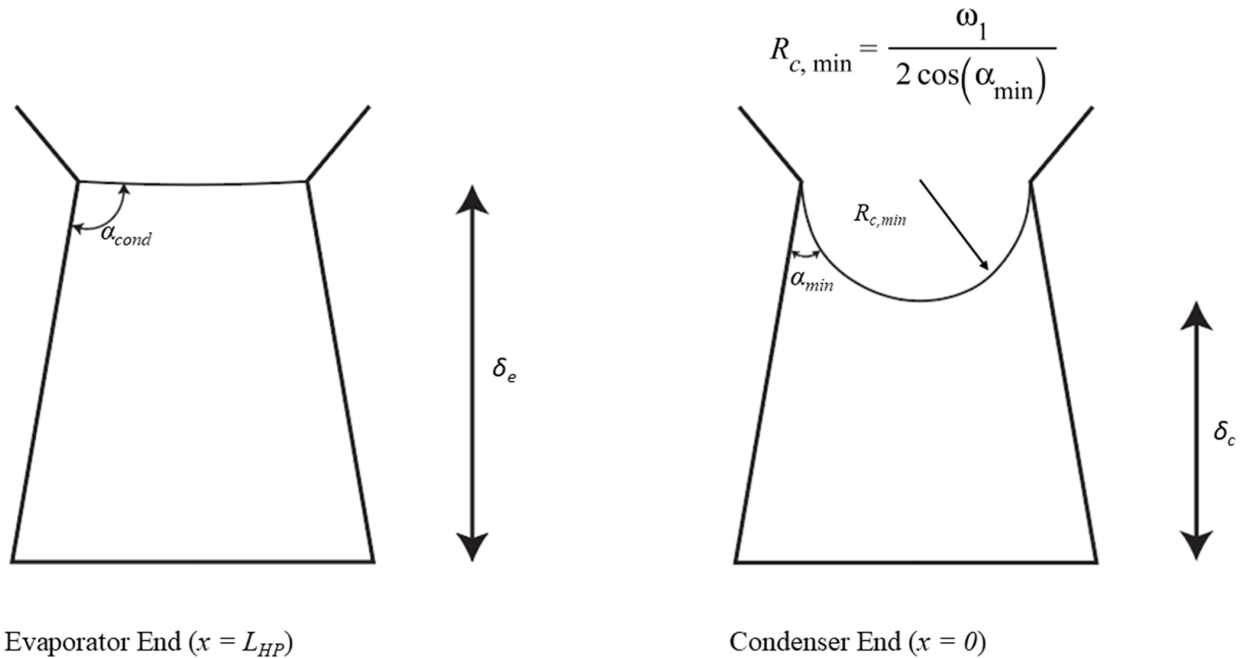


FIG. 5: Thermal resistances inside the axial grooves in the evaporator and condenser sections

Hence, to find the Q_{max} , the initial value of Q_{in} is taken as 1 W, with $\Delta Q = 1$ W. If the limiting condition of $R_{c,min}$ is not reached, then the value of Q_{in} is incremented by ΔQ and the model is run again until the value of Q_{max} is obtained. The flowchart given in Fig. 6 explains this solution algorithm.

For our study, to observe the effects of different angles of bend in the AGHPs at different operating temperatures, we ran the numerical model across the range of -60°C to 100°C (also considered to be the operating range for ammonia-based heat pipes for space applications) with increments of 20°C . For each operating temperature, we analyze the effect of bend at bend angles varying from 0° (conventional straight heat pipes) to 180° (U-bend heat pipes) with increments of 30° . Hence, upon running the model for 7 bend angles and 9 operating temperatures (totaling 63 configurations), we analyze the observations and comment on the trends observed in the maximum rate of heat transfer (Q_{max}) as well as the total thermal resistance (R_{Total}) of the heat pipe.

2.4 Validation of the Mathematical Model

To validate the current mathematical model, it is first implemented in the previous studies conducted by Desai et al. (2019) and Anand (2019) on conventional AGHPs, and the results are compared with their experimental results.

Upon comparison with the results obtained by Desai et al. (2019), it is observed that the model accurately predicts (within 0.5% deviation) the values of Q_{max} reported when considered with $\pm 1^\circ$ of variation in limiting contact angle value of ammonia with aluminum. This consideration can be established in the limiting contact angle value due to its

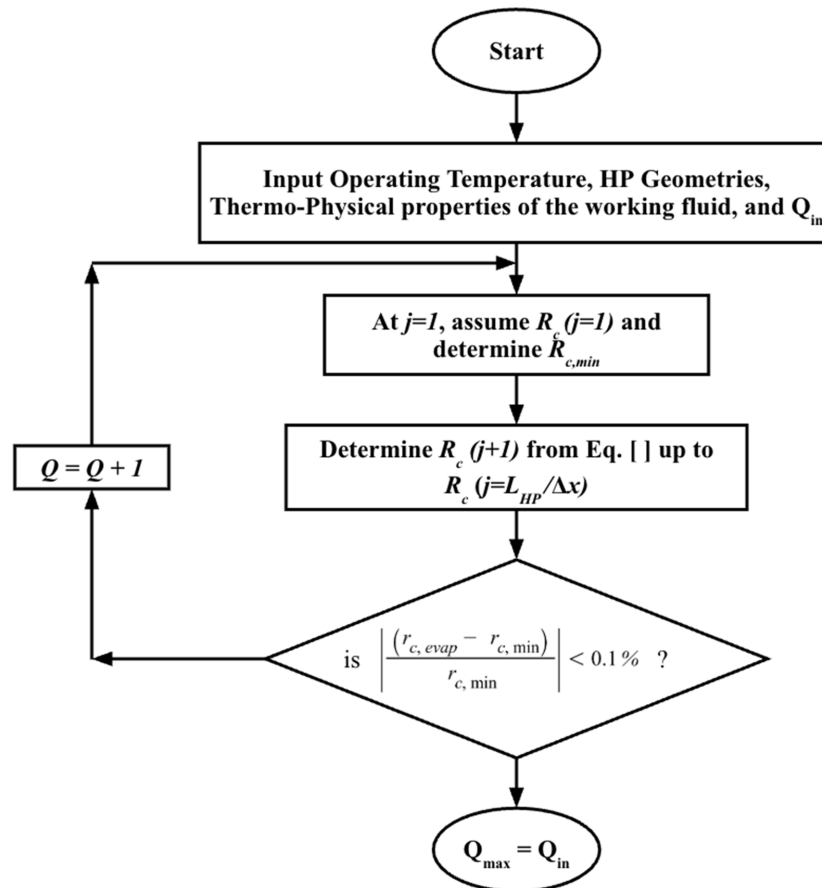


FIG. 6: Flowchart for the solution algorithm

dependence on several other fluids, and the solid thermophysical properties and can alter its value to a certain extent at different temperatures. This can be explained by the fact that the surface wetting property of any fluid is dynamic with the environmental conditions, which include the temperature as well.

Upon implementing the numerical model for aluminum–methane AGHP studied by Anand (2019) (Fig. 7), the model predicts the values of heat transport capability with nearly 2% error in the temperature range above 110 K. At 100 K, the prediction shows a 12% deviation, which may be possible due to the drastic changes in the thermophysical properties of methane at extremely low temperatures.

3. RESULTS AND DISCUSSION

The numerical model was run for the operating temperatures from -60°C to 100°C with an increment of 20°C and bent in the middle of the adiabatic section, with bend angles varying from 0° (straight heat pipe) to 180° (U-bend heat pipe) with 30° increments. The results obtained from the abovementioned 63 combinations (9 operating temperatures and 7 bend configurations) were then analyzed to predict the behavior of the AGHP in terms of its maximum heat transfer capacity (Q_{max}) and total thermal resistance (R_{total}).

It is seen that at an operating temperature of -20°C , the maximum heat transfer capacity is the highest when compared to other operating temperatures at all angles of bend. This also goes by the value of the figure of merit for ammonia as a working fluid in heat pipes. Due to this, the plots of variations of different physical properties inside the heat pipe are also the most defined when observed at -20°C . The trend for all other operating temperatures remains similar to that at -20°C .

As shown by Fig. 8, it is observed that at -20°C , there is a decrement in the value of the maximum heat transfer capacity when there is an increment in the angle of bend; that is, at any operating temperature, the configuration with no-bend (0° bend angle) shows the highest Q_{max} (131 W), while the configuration with U-bend (180° bend angle) shows the least value of Q_{max} (93 W). Furthermore, it is observed that as the bend angle is increased, the total thermal

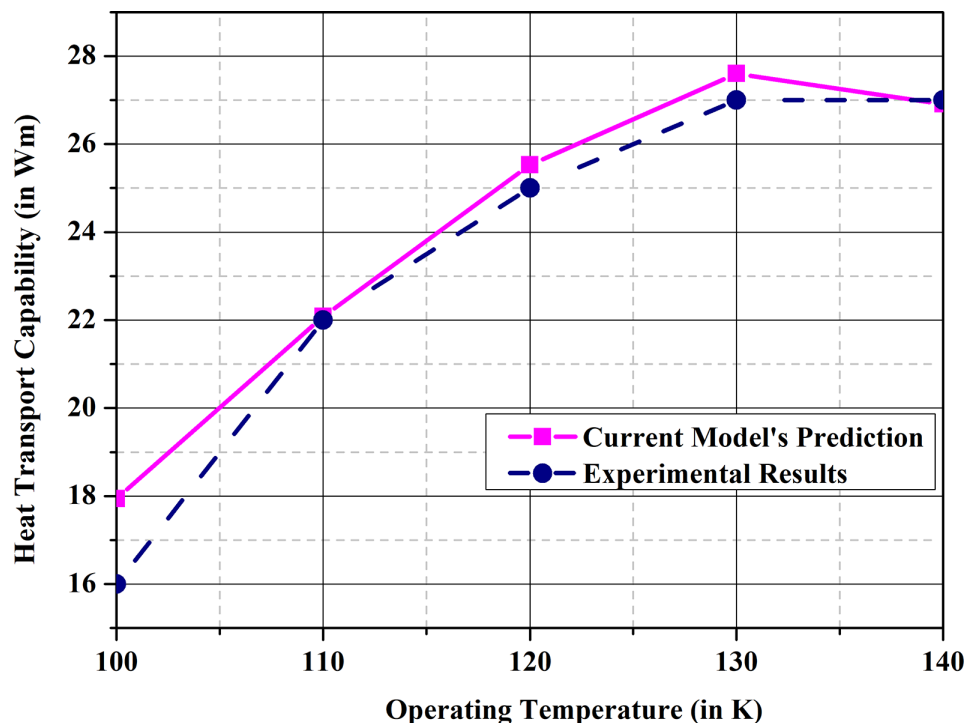


FIG. 7: Prediction against the experimentally reported results (Anand, 2019) of heat transport capability

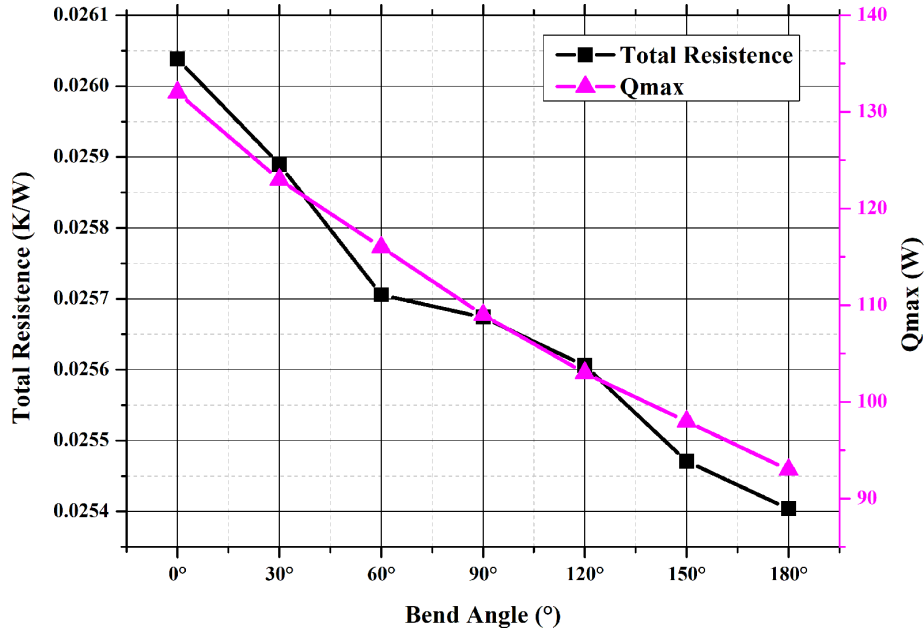


FIG. 8: Values of $Q_{\max}$ and R_{Total} obtained at -20°C at different bend angles

resistance decreases in the majority of scenarios. Hence, for a particular operating temperature, the configuration with no-bend (0° bend angle) shows the highest value of R_{Total} , while the configuration with U-bend (180° bend angle) shows the lowest value of R_{Total} .

The behaviors of different working fluid thermophysical properties, like pressure head, velocity, and mass flow rate, are observed across the length of the AGHP in the following subsections and utilized to justify these trends in the value of $Q_{\max}$. Similarly, the variation of the vapor–liquid meniscus radius (R_c) and the groove liquid-layer thickness (δ) are inferred to explain the behavior of the total thermal resistance.

3.1 Effect of Bending on the Pressure Drop across the Heat Pipe

When observed at the -20°C operating temperature, it is seen that the total pressure drops remain the same in both phases (for all bend angles) along the heat pipe length (as depicted by Fig. 9). This helps reestablish the fact that the overall pressure drop across the heat pipe is a result of the capillary action that its wick can provide, which is a function of its surface energy and groove dimensions. Hence, it supports the fact that at any operating temperature, the maximum capillary pressure that can be generated by a given heat pipe is constant, irrespective of the existence or extent of bends in the pipe.

At any given instant, the overall pressure drop occurring in the heat pipe is equal to the capillary pressure that its wick structure generates. As the driving force of the heat pipe, all individual pressure drops (namely P_p , P_v , P_{gravity} , and P_{bend} when minor losses are ignored) add up to the capillary pressure that the heat pipe wick develops.

$$\Delta P_{\text{Total}} = \Delta P_l + \Delta P_v + \Delta P_{\text{gravity}} + \Delta P_{\text{bend}} \quad (44)$$

Since the study is performed in microgravity conditions,

$$\Delta P_{\text{Total}} = \Delta P_l + \Delta P_v + \Delta P_{\text{bend}} \quad (45)$$

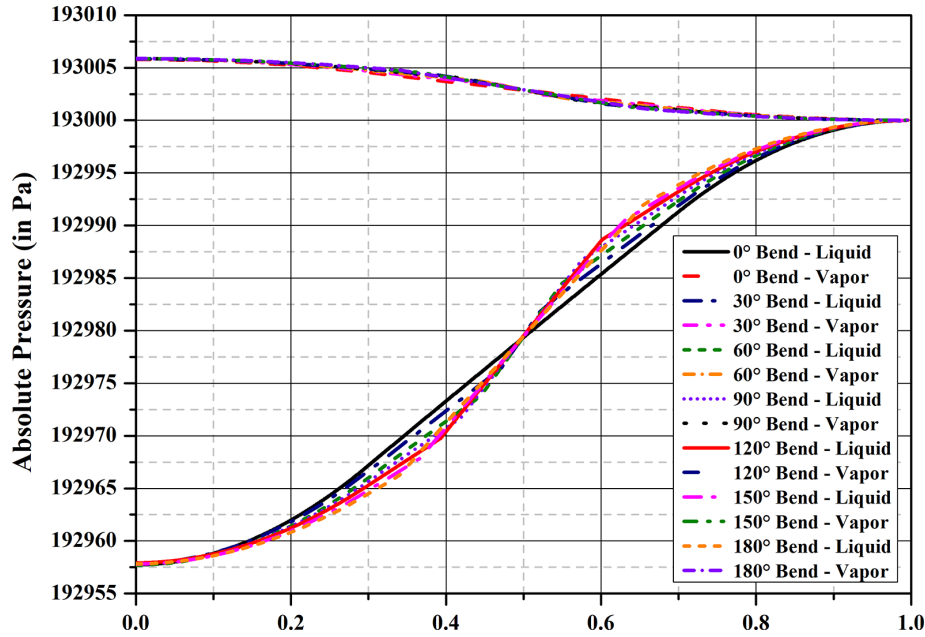


FIG. 9: Pressure variation inside the heat pipe with different bend angles at -20°C operating temperature

As mentioned earlier, the maximum capillary pressure is fixed, and at Q_{max} it is equal to

$$\Delta P_{Total} = \Delta P_{capillary} = \frac{2 \cdot \sigma}{R_{c,min}} \quad (46)$$

where $R_{c,min}$ is fixed for given groove dimensions and at a given operating temperature. Hence, even when a fraction of the total pressure head is lost at the bend due to loss of energy while changing the direction of motion and the formation of eddies, the overall pressure drop that occurs inside the heat pipe remains the same. Since the total pressure drop remains the same for all bend angles, the higher the value of the pressure drop around the bend, the lesser the pressure drop allotted to the liquid and vapor pressure drops across the heat pipe. Hence, with the increase in bend angle, and therefore the increase in the drop in pressure across the bend, the capability of the heat pipe to transport the fluid (both liquid and vapor phases) decreases, which results in a lower Q_{max} .

The nature of the pressure curve for each bend angle varies from one another to a great extent. When we compare a heat pipe with bends with a straight heat pipe of identical properties, due to less pressure being distributed for P_l and P_v for the former heat pipe, the pressure gradients of liquid pressure and vapor pressure are seen to be more gradual (less steep) as we move along the axis of the heat pipe except when moving across the bend. As seen in Figs. 10 and 11, it is observed that the higher the angle of bend, the less steep the gradient of the pressure curves (as less pressure will be allotted to P_l and P_v). As we approach the bend, it is seen that the higher the bend angle, the steeper the pressure gradient in pressure curves of both phases along the entire length of the bend. Finally, this peculiar behavior of the liquid and vapor pressure curves allows them to converge with the same value of total pressure drop for every bend angle.

3.2 Effect of Bending on the Fluid Velocities and Mass Flow Rates across the Heat Pipe

It is observed that as the bend angle is increased, the respective liquid (Fig. 12) and vapor velocities (Fig. 13), and hence the mass flow rates (Fig. 14) of the working fluid, decrease uniformly. This observation goes per our inference

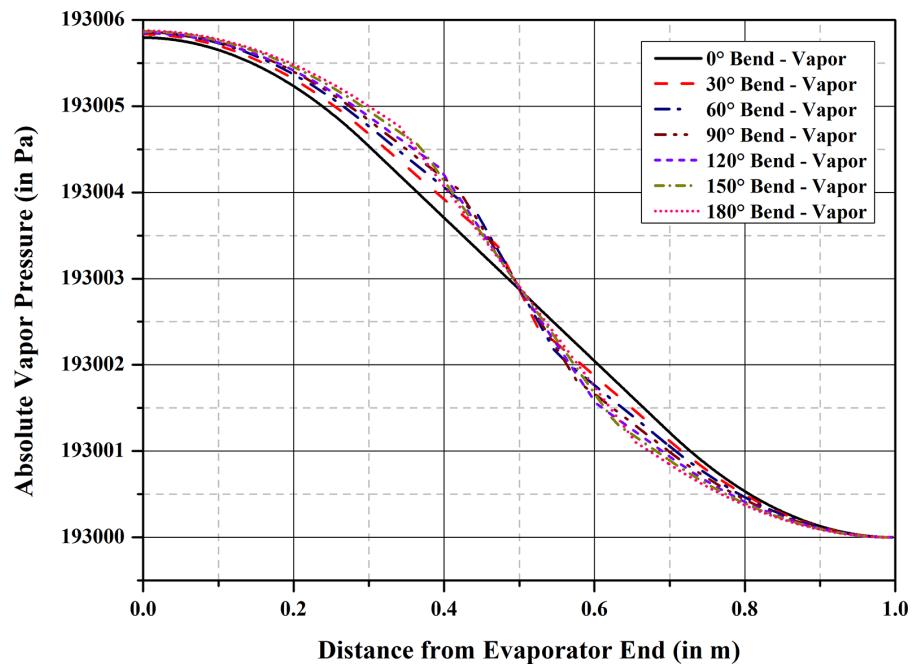


FIG. 10: Pressure variation inside the heat pipe with different bend angles at -20°C operating temperature

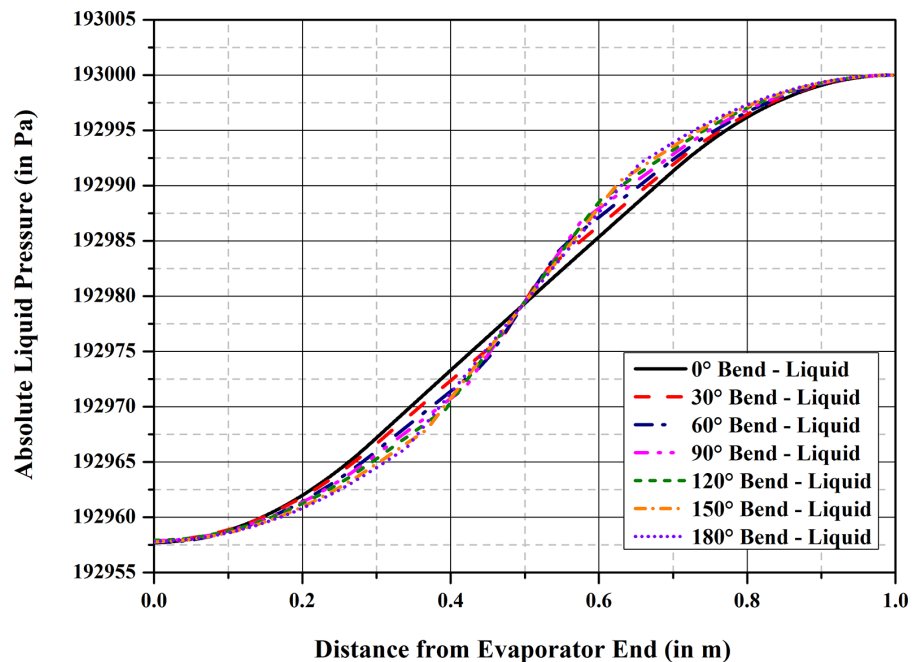


FIG. 11: Liquid pressure variation inside the heat pipe with different bend angles at -20°C operating temperature

from the previous section; that is, as more and more pressure head is lost at the bend, the less the pressure head allotted for the working fluid transportation in the liquid and vapor phases will be. Since the liquid pressure drop (P_l) will be responsible for pushing the liquid from the condenser end to the evaporator end, being assigned a smaller fraction

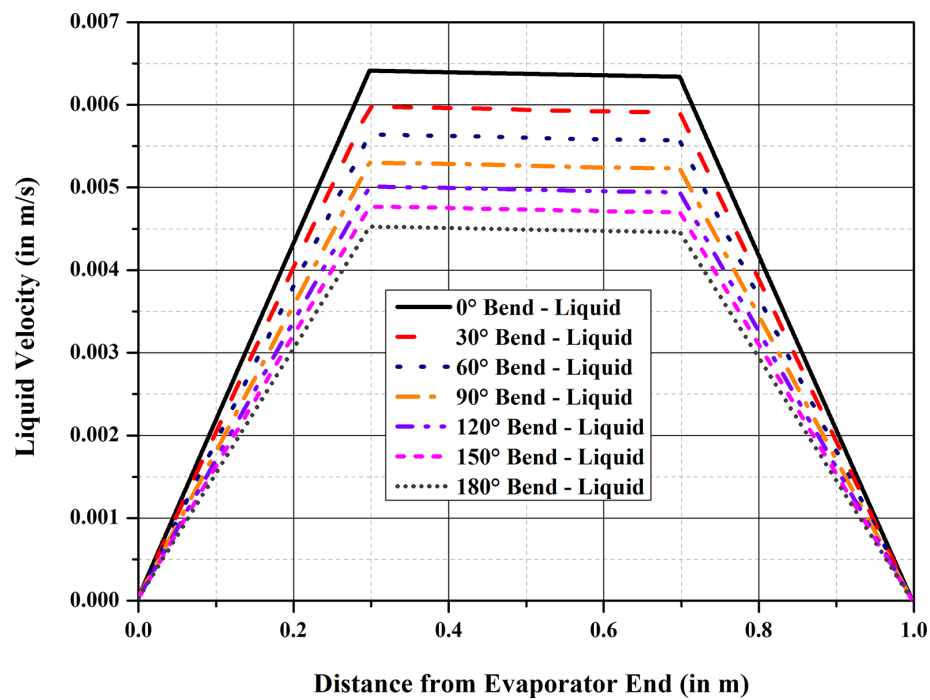


FIG. 12: Liquid velocity variation inside the heat pipe with different bend angles at -20°C operating temperature

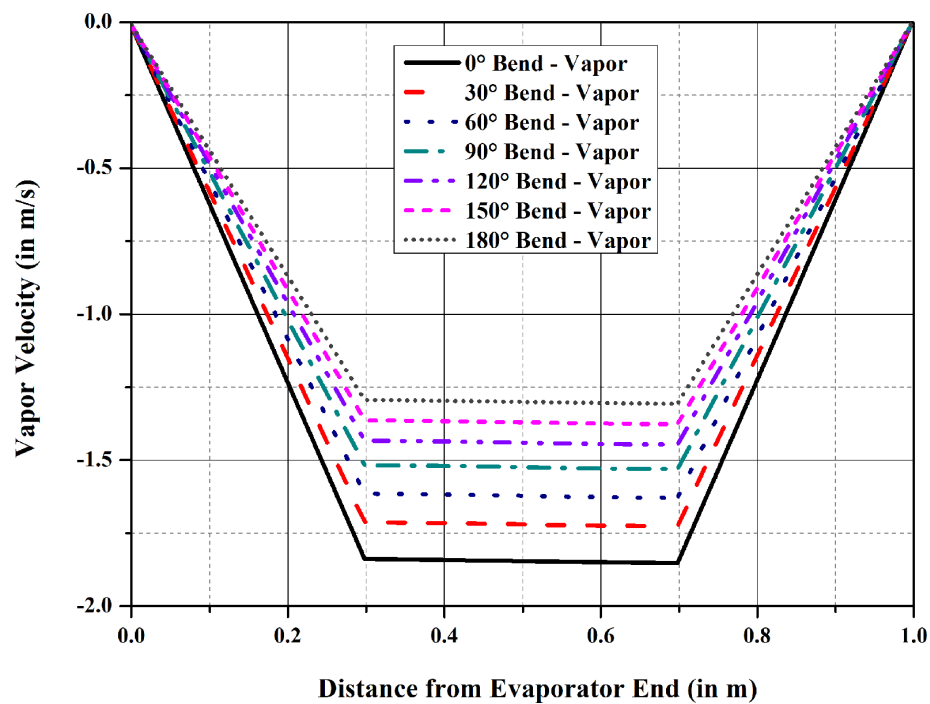


FIG. 13: Vapor velocity variation inside the heat pipe with different bend angles at -20°C operating temperature

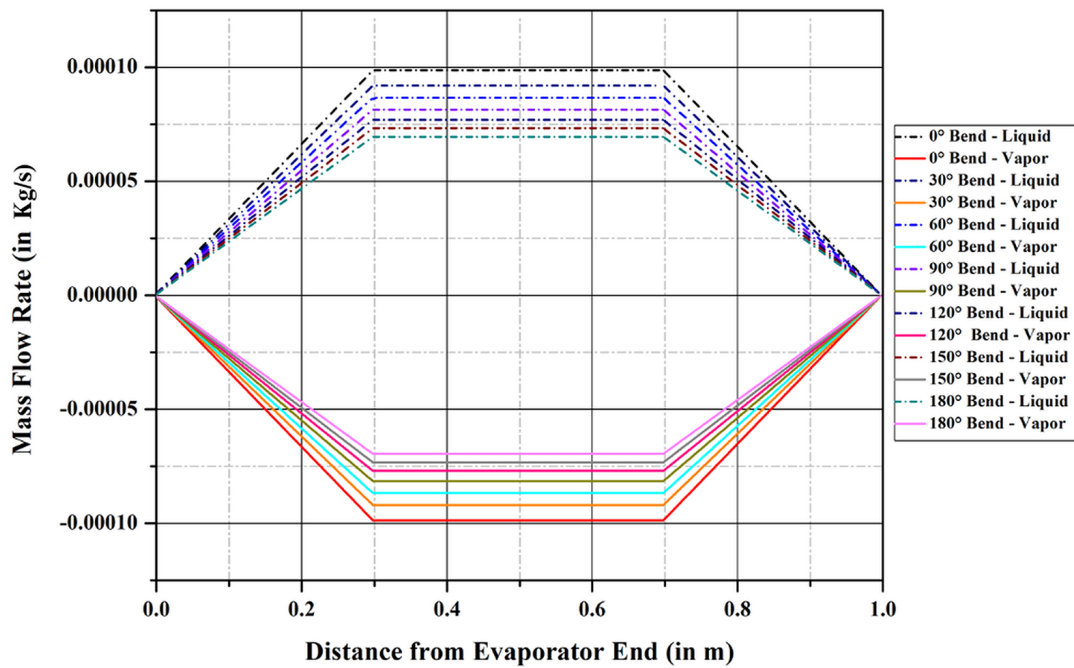


FIG. 14: Liquid and vapor mass flow-rate variation inside the heat pipe with different bend angles at -20°C operating temperature

of the total pressure will result in a lower liquid velocity. Similarly, the decrease in pressure head assigned to vapor pressure drop (P_v) will result in a decreased vapor velocity inside the vapor core.

Since the heat pipe transfers (or rather transports) heat using the principle of latent heat of vaporization, having a lower possible value of the maximum mass flow rate (at the respective Q_{max}) due to a higher bend angle will result in a lower rate of heat transfer. Hence, Q_{max} will be lower for the bent heat pipe when compared to an identical straight heat pipe (except for the bend).

3.3 Effect of Bending on the Maximum Heat Transfer Capacity (Q_{max})

When the bend angle is increased, the value of Q_{max} at any given operating temperature decreases (as shown in Fig. 15). This effect is more prominent in the middle of the temperature range than it is at the extremes. Also, it should be noted that the observed trends at any given bend angle fall in agreement with the trend of the figure of merit value of ammonia as a working fluid in heat pipes within our range of operating temperatures. Anand (2021) also stated the T_{opt} or optimal temperature, for heat pipe operation of ammonia as a working fluid as -20°C , which is backed by our observations on the variation of Q_{max} (maxima of Q_{max} curves for all bend angles at -20°C) using the current numerical model.

As we discussed in Sections 3.1 and 3.2, the tendencies observed in the variations in pressure drops, velocities, and mass flow rates in both the phases can be used very well to explain the trend that we observe in the value of Q_{max} .

If we compare the cases of no-bend (0° bend angle) and U-bend (180° bend angle) at the operating temperature of -20°C (where total capillary pressure is estimated to be about 48 Pa), we notice a decrease in the value of Q_{max} from 131 W at 0° bend to 93 W at 180° bend (nearly 29% decrease) due to the loss of pressure head at the bend, which was equal to nearly 14 Pa (29% of the total capillary pressure). Due to this, the remaining pressure head of 34 Pa could provide only for the fluid mass flow rate of $6.95\text{E-}05 \text{ Kg}\cdot\text{m/s}$, in place of $9.86\text{E-}05 \text{ Kg}\cdot\text{m/s}$ (29% decrease), which was possible at a 0° bend.

These similarities in the percentage decrease for values of Q_{max} , pressure head, and mass flow rate help us establish the dependence of Q_{max} on these physical properties.

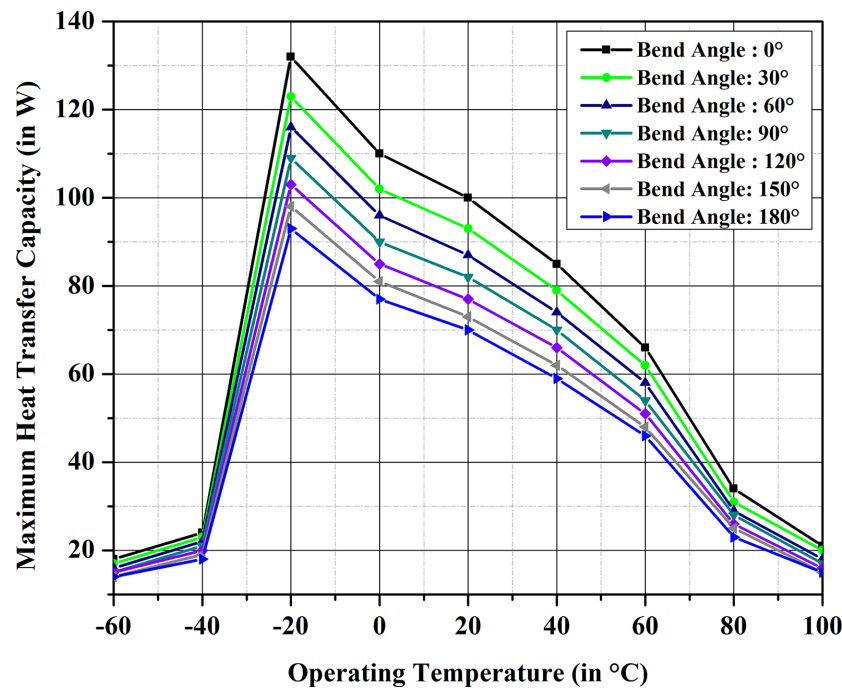


FIG. 15: Variation of $Q_{\max}$ across the entire operating range at different bend angles

3.4 Effect of Bending on the Liquid–Vapor Meniscus

As previously discussed, the functioning and heat transfer performance of any AGHP depend on the liquid–vapor interaction film that separates the vapors in the vapor core from the liquid in the wick of the heat pipe. The nature of the arrangement of liquid inside the axial grooves along the heat pipe axis can determine a lot about the thermal resistance offered by the AGHP. Most importantly, it also determines at what value of the heat input the dry-out will start to occur in the heat pipe.

It is observed in Fig. 16 that for a 0° bend angle, there is a steep decrease in the radius of the meniscus near the condenser end. As we start moving toward the evaporator end, the slope of variation of the radius of the meniscus keeps decreasing until it becomes almost flat at the evaporator end. On the other hand, in the case of 180° bend-angle (as it depicts the changes in the most pronounced manner), it is seen that the slope is slightly less steep than the curve of 0° bend-angle owing to the reduced pressure head responsible for fluid transport. As we reach the bend, a sharp reduction in the slope of R_c variation is observed due to the loss of pressure head throughout the bend. We observe that at the center of the bend, the value of R_c becomes equal for all bend angles and then becomes less and less than that of 0° for all bend angles until they converge again at the evaporator end. As discussed before, this convergence takes place because the value of $R_{c,min}$ is the same for all bend angles.

This transition is seen more evidently when we come across the variation of groove liquid-layer thickness (δ), which is the height up to which the groove is filled at a given point along the heat pipe's axis (Fig. 17). It must be noted that the positioning of a bend in the middle of the adiabatic region results in additional liquid accumulation near the condenser region and reduced liquid accumulation near the evaporator region when compared to a straight heat pipe.

3.5 Effect of Bending on the Total Thermal Resistance (R_{Total})

When we come across the trend followed by R_{evap} , R_{cond} , R_{adia} , and R_{Total} for various bend angles at -20°C operating temperature, we find that R_{evap} is the main contributor in the R_{Total} and is much more influential in determining the R_{Total} than R_{cond} (10^{-1} times the magnitude of R_{evap}) and R_{adia} (10^{-4} times the magnitude of R_{evap}).

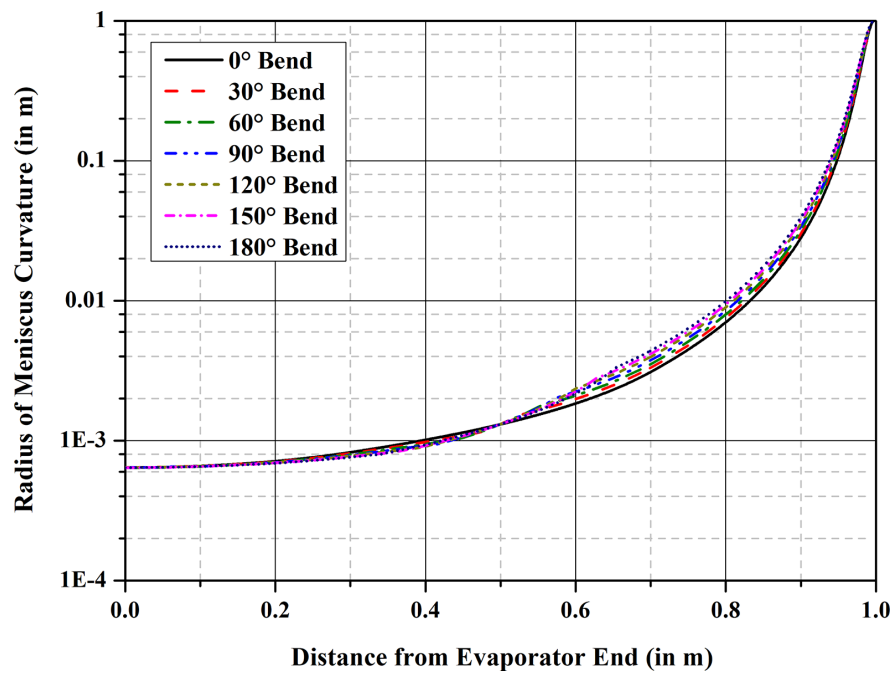


FIG. 16: Liquid–vapor film meniscus radius variation with different bend angles at -20°C operating temperature

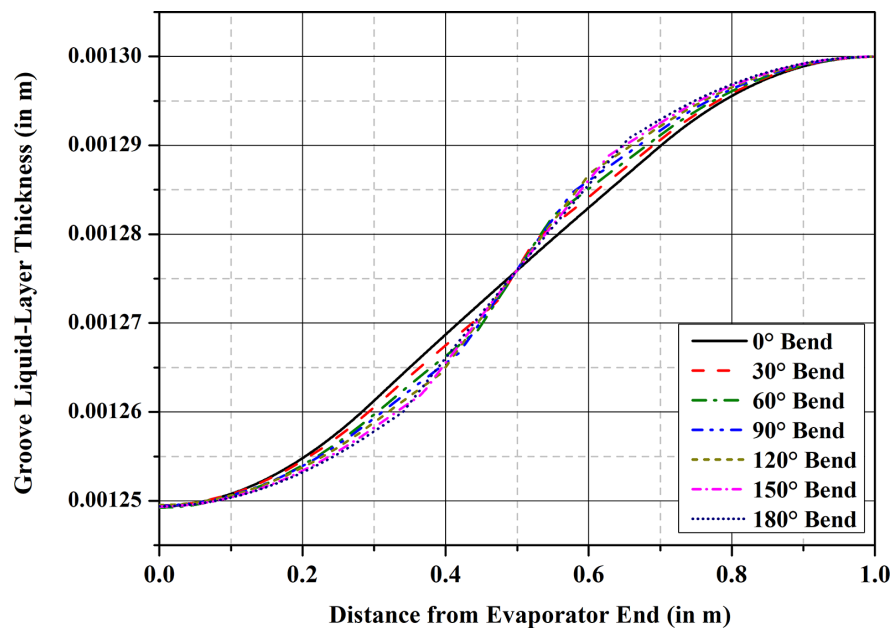


FIG. 17: Groove liquid layer thickness variation with different bend angles at -20°C operating temperature

Hence, when we look closely at the trends of R_{Total} and R_{evap} , we observe that their values gradually decrease as we increase the bend angle (Fig. 18).

Arab and Abbas (2014) established in the past that the micro (thin film) region of the heat pipe serves as the route for the transmission of nearly 80% of the total heat input. In addition to this, there will be negligible changes

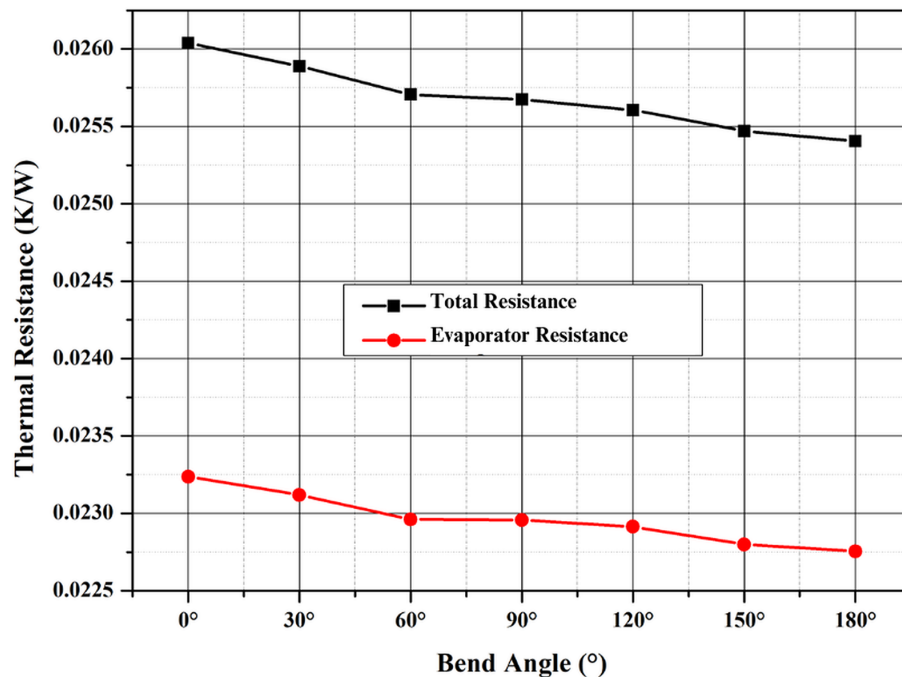


FIG. 18: Heat pipe total resistance and evaporator resistance at -20°C operating temperature for different bend angles

in the resistance through the metal body of the heat pipe due to the presence of the bend, as the resistance of a metal body depends on the length and area of the cross section of the body, which will have negligible change upon bending.

Along the same lines, the trend observed in R_{evap} and hence R_{Total} can be very well explained by the variation of R_c and the variation of groove liquid-layer thickness (δ). As shown in Fig. 17, it can be observed that the quantity of liquid accumulated away from the evaporator end increases as the bend angle increases from 0° to 180° , reducing the microregion within the evaporator section, which is primarily responsible for the total thermal resistance offered by the AGHP. Furthermore, such a high dependence on the thickness of the liquid film inside the evaporator section can also be explained by the fact that the conductivity of ammonia (approx. 0.300 W/m-K) is very low when compared to aluminum (approx. 240 W/m-K). Since most of the heat is provided to the fluid in the evaporator section (as the boiling occurs here), the liquid film thickness in the evaporator section particularly plays a major role in the determination of R_{Total} .

The abovementioned trend is most pronounced when observed at -20°C operating temperature and can be seen for the temperature range from -20°C to 60°C (as depicted in Fig. 15).

However, as we go beyond this range to the extreme temperatures (below -20°C and above 60°C), the above explanation does not stand completely true. This discrepancy can arise when the thermophysical properties of the working fluid start changing drastically as we approach the end of its working temperature range. As a general trend, it is also observed that at -40°C , the R_{Total} is found to be the lowest for all bend angles.

The thermal resistance offered by a heat pipe, when used for space applications, plays a crucial role in designing the thermal management system of the spacecraft. A higher thermal resistance can prove problematic as it can cause issues like the reduction in the heat transfer capacity of the system, or an increase in the overall weight of the payload (since a greater number of heat pipes will be required to attain the appropriate total resistance). Additionally, a higher resistance will cause more load on the radiator, lowering its temperature. To counter this, a radiator with a larger surface area will be necessary, which will result in a heavier payload.

Hence, optimizing the heat pipe design for its application and thermal (heat transfer) requirements involving Q_{max} and R_{Total} is of grave importance. Based on our analysis, we can be sure of two trends. First, with an increase in the

angle of bend, the maximum heat transfer capacity at a particular operating temperature decreases. Second, with the increase in the angle of bend, the total thermal resistance at a particular operating temperature also decreases. Hence, for applications where one or more angles of the bend will be possible, the heat pipe design and its bend angle can be optimized such that the Q_{max} can be maximized while minimizing the R_{Total} .

4. CONCLUSION

This study intended to develop a mathematical model to understand the performance of bent heat pipes of any given geometrical and physical specifications and to analyze how the bends will affect their maximum rate of heat transfer and total thermal resistance. To implement the effect of bending, the physics was derived from studies performed on the analysis of the loss of pressure head in large steam pipes. The empirical formulae were hence adopted into the mathematical model for straight heat pipes working on the principle of disruption of the liquid–vapor interaction film, by calculating the pressure head loss gradient occurring inside the vapor core carrying gaseous ammonia and in the axial grooves carrying liquid ammonia. Upon validation of the model using existing experimental studies, it could accurately predict the heat transfer performance of AGHPs with bends under microgravity conditions for space applications.

After analyzing the results obtained from 63 different combinations of the operating temperatures and bend angles, it was observed that:

- As the bend angle increases, the maximum rate of heat transfer (Q_{max}) at a particular operating temperature decreases in value. This is supported by the increase in pressure drop seen at the bend as the bend angle is increased. Since there is a higher pressure drop allotted at the bend, the pressure head that remains for the flow of fluids reduces, reducing the mass flow rate of the fluid and the maximum rate of heat transfer in direct proportion.
- The values of Q_{max} obtained at any bend angle fall in line with the trend seen in the figure of merit of ammonia at different operating temperatures. The value of Q_{max} gradually decreases as the bend angle is increased, throughout the operating temperature range discussed here (-60°C to 100°C). The reduction in the value of Q_{max} at any operating temperature is seen to be proportional to the value of Q_{max} observed at a 0° bend angle (highest difference in Q_{max} at -20°C , and lowest difference in Q_{max} at -60°C).
- The total thermal resistance (R_{Total}) at a given operating temperature is also observed to decrease in value (in most cases) as the bend angle is increased. This observation is supported by the decrease seen in the liquid film thickness (microregion) in the evaporator section as the bend angle increases.
- The trend in R_{Total} mentioned above is seen clearly for the range of -20°C to 60°C . Before and after this temperature range, the trend is not very pronounced, which can be the result of different thermophysical properties of ammonia at extreme temperature values. It is observed that the thermophysical conditions at -40°C offer the best conditions concerning R_{Total} and hence the lowest value for every bend configuration.

This numerical model was also developed with the objective of being used as part of the design optimization algorithms for the thermal management of spacecraft payloads. This model has the potential to help toward obtaining a more accurate prediction of the heat transfer characteristics of bent heat pipes, and thus allow for huge savings in costs incurred in prototyping heat pipes of several bend angles and analyzing their performance experimentally until the required specifications (in terms of its heat transfer characteristics and the geometric feasibility within the system) are met. This model will also help us determine the heat transfer performance of flexible heat pipes, and hence pave the way for their design and development.

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